



ORACLE
NetSuite

SuiteCloud SDK Overview



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SuiteCloud SDK Overview

Important: Starting February 2025, SuiteCloud SDK version 24.1, the last version that supports Token-Based Authentication (TBA) and OAuth 1.0, will no longer be available for new installations. If you're still using older versions of the SuiteCloud SDK tools, you should upgrade to the latest version as soon as possible.

SuiteCloud Software Development Kit (SuiteCloud SDK) are the tools that allow you to customize accounts through SuiteCloud Development Framework (SDF).

Use the SDK tools to work with an IDE plug-in or an extension to develop SuiteCloud projects. If you're familiar with an IDE or a code editor of your choice, use SuiteCloud Command Line Interface (SuiteCloud CLI) through a terminal to develop SuiteCloud projects.

The following tools are available:

- To use a plug-in developed for a specific IDE or to use an extension, see [SuiteCloud Plug-ins and Extensions](#).
- To use a CLI, see [SuiteCloud CLIs](#).

For a list of the available features for each tool, see [Overview of Features by Tool](#).

Important: Before using any of the SuiteCloud SDK tools, ensure that your role fulfills all the requirements. For more information, see the help topic [Assigning the Developer Role \(Administrator Only\)](#).

Sample projects

If you're not familiar with how SuiteCloud projects work, you can download sample projects that are ready to be used. You can modify the SuiteScript files, SDF custom objects, and all the files in the projects as you see fit. For more information, see the help topic [SuiteCloud Customization Tutorials](#).

Important: To work with these SuiteCloud projects in SuiteCloud CLI for Node.js, first run `suitecloud project:create` to create a SuiteApp project. Then, download a ZIP file from the Help Center and extract its contents in the `src` folder of the SuiteApp you created.

SuiteCloud Plug-ins and Extensions

Use these tools to work with an IDE plug-in or extension and take advantage of an intuitive user interface to develop SuiteCloud projects.

SuiteCloud IDE plug-ins and extensions, designed for NetSuite platform development, provide object XML definitions used in SuiteCloud projects. You can interact with File Cabinet files and SDF custom objects, create projects and SuiteScript files, and deploy and validate SuiteCloud projects. You can easily do all these actions across different accounts thanks to straightforward account management.

The following IDE plug-ins and extensions are available:

- [SuiteCloud Extension for Visual Studio Code \(VS Code\)](#)

- [SuiteCloud IDE Plug-in for WebStorm](#)
-  [SuiteCloud IDE for Eclipse](#)

 **Note:** Due to Eclipse not receiving any further updates, the documentation is only available in PDF format.

SuiteCloud Extension for Visual Studio Code (VS Code)

If you prefer a JavaScript and TypeScript-based extension with built-in support for Node.js, see the help topic [SuiteCloud Extension for Visual Studio Code Overview](#).

Also, to see how to use the basic features of SuiteCloud Extension for VS Code, watch the following video:

 [SuiteCloud Extension for Visual Studio Code \(VS Code\)](#)

SuiteCloud IDE Plug-in for WebStorm

If you prefer a JavaScript and TypeScript-based IDE, see the help topic [SuiteCloud IDE Plug-in for WebStorm Overview](#).

SuiteCloud IDE Plug-in for Eclipse

 **Note:** As of NetSuite 2022.2, there will be no new releases for SuiteCloud IDE Plug-in for Eclipse. SuiteCloud IDE Plug-in for Eclipse remains available, but no further features will be added, and issues will not be fixed. For our latest development tools, consider transitioning to SuiteCloud Extension for Visual Studio Code, SuiteCloud IDE Plug-In for WebStorm, or SuiteCloud CLI for Node.js.

If you prefer a Java-based IDE, see  [SuiteCloud IDE for Eclipse Guide](#).

Due to Eclipse not receiving any further updates, the documentation is only available in PDF format.

SuiteCloud CLIs

If you're familiar with an IDE or a code editor of your choice, you can use SuiteCloud Command Line Interface (SuiteCloud CLI) through a terminal to develop SuiteCloud projects.

The following CLIs are available:

- [SuiteCloud CLI for Java](#)
- [SuiteCloud CLI for Node.js](#)

SuiteCloud CLI for Node.js

This CLI is interactive and guides you through every step of connecting your local project to your account. You can also use it for integrations and automations.

To learn more about SuiteCloud CLI, see the help topic [SuiteCloud CLI for Node.js Guide](#).

SuiteCloud CLI for Java

This CLI is built on Java and is optimized for integrations and automations.

To learn more about SuiteCloud CLI, see the help topic [SuiteCloud CLI for Java Guide](#).

Overview of Features by Tool

Here's a list of the available features for each tool in the table below.

Note: The information in the table below is for the latest version of each tool.

Feature	SuiteCloud Extension for Visual Studio Code (VS Code)	SuiteCloud IDE Plug-in for WebStorm	SuiteCloud CLI for Node.js	SuiteCloud CLI for Java
Authenticate (browser-based and machine-to-machine account set up)	Available	Available	Available	Available
Manage authentication IDs	Available	Available	Available	Available
Create project (for ACPs and SuiteApps)	Available	Available	Available	Available
Unit testing with Jest	Available	Available*	Available	Not available
Create object	Not available	Available	Not applicable	Not applicable
Import object	Available	Available	Available	Available
Import custom record instances	Available	Available	Available	Available
Update object	Available	Available	Available	Available
List object	Available	Available	Available	Available
Compare object	Available	Available	Not applicable	Not applicable
Create file	Available	Available	Available	Available
Import file	Available	Available	Available	Available
Upload file	Available	Available	Available	Available
List file	Available	Available	Available	Available
Compare file	Available	Available	Not applicable	Not applicable
Upload folder	Not available	Available	Not available	Available
Add dependencies	Available	Available	Available	Available
Code completion (for SuiteScript files)	Not available	Available	Not applicable	Not applicable

Feature	SuiteCloud Extension for Visual Studio Code (VS Code)	SuiteCloud IDE Plug-in for WebStorm	SuiteCloud CLI for Node.js	SuiteCloud CLI for Java
Dry run/Preview	Not available	Available*	Available	Available
Run debug configuration	Not available	Available	Not available	Not available
Package project	Not available	Available	Available	Available
Validate locally	Available	Not available	Available	Available
Validate server-side	Available	Available	Available	Available
Deploy	Available	Available	Available	Available
Account specific values (for ACPs)	Available	Available	Available	Available
Create installation preferences file (for SuiteApps)	Not available	Available	Not applicable	Not applicable
Apply installation preferences (for SuiteApps)	Available	Available	Available	Available

*Achievable through run debug.

OAuth 2.0 Authentication for SuiteCloud SDK

SuiteCloud SDK uses the OAuth 2.0 protocol for authorization. To use this protocol, you need to have the OAuth 2.0 feature enabled in the account.

With OAuth 2.0, you can choose between two options when authenticating with any of the SuiteCloud SDK tools:

- **Browser-based authentication:** This option opens a new window or tab in your default browser, so you can authenticate your account-role combination. You'll need to interact with the browser for this option.

The first time you use browser-based authentication, you are asked to click **Allow** in the browser. For the subsequent occasions, you are asked to click **Continue**.



Warning: With browser-based authentication, you'll need to reauthorize each time your authorization details expire. You'll get a notification, and the browser will open for you to reauthorize.

- **Machine-to-machine authentication:** This option uses digital certificates and needs some setup before you can authenticate your account-role combination. When it's set up, you don't need any user interaction.

This option is meant to be used for CI environments.

With machine-to-machine authentication, you need to enter the following details:

- **Account ID:** References the NetSuite account ID.
- **Certificate ID:** References the certificate ID you get from NetSuite after uploading the public key file.
- **Private key file:** References the local path to the private key file that matches the certificate in NetSuite. For example: d:/path/private-key.pem.



Important: To get the **certificate ID**, you need to generate an **RSA certificate**. The RSA certificate has two parts: the **public key** and the **private key**.

Upload the public key in the OAuth 2.0 Client Credentials Setup by choosing **SuiteCloud Development Integration** in the **Application** field and selecting the entity and role to map. This will generate a new row with the certificate ID value.

The private key path value is the location where the private key is saved.

For more information about how to generate the **RSA certificate**, see the help topic [Certificate Conditions](#).

For more information about how to upload the public key, follow the steps in [OAuth 2.0 Client Credentials Setup](#).



Note: If you're authenticating for the first time, use the browser-based authentication to get the integration record installed in your account.

To authenticate using SuiteCloud Extension for Visual Studio Code, see the help topic [Adding an Account in SuiteCloud Extension for Visual Studio Code](#).

To authenticate using SuiteCloud IDE Plug-in for WebStorm, see the help topic [Adding an Account in SuiteCloud IDE Plug-in for WebStorm](#).

To authenticate using SuiteCloud CLI for Node.js, see the help topic [account:setup](#) or [account:setup:ci](#).

To authenticate using SuiteCloud CLI for Java, see the help topic [authenticate](#) or [authenticateci](#).

Secure Credentials Storage for SuiteCloud SDK

Starting in 2025.1, SuiteCloud SDK uses Public-Key Cryptography Standards #12 (PKCS#12) to generate a password-protected credentials file.

When you authenticate for the first time using SuiteCloud SDK tools, the PKCS#12 credentials file is created and stored on your local machine in the `.suitecloud-sdk` folder in your home directory. This file contains encrypted authentication data, so you don't have to authenticate again in future sessions. The credentials file name depends on the authentication option you used: `credentials_browser_based.p12` for browser-based authentication or `credentials_ci.p12` for machine-to-machine authentication.



Important: Browser-based authentication and Machine-to-machine authentication use different credential files to store authentication data. You can set up your environment to use only one authentication option at a time.

SuiteCloud SDK checks for specific environment variables and the availability of secure storage to determine if you've got the right setup for the authentication option that you're trying to run. For more information, see [Execution Context for Secure Credentials Storage](#).

To keep your credentials file safe, SuiteCloud SDK encrypts it with a password or passkey:

- For browser-based authentication, SuiteCloud SDK auto-generates the passkey and updates it regularly. This passkey is stored in your machine's secure storage service (like Credential Manager for Windows, Keyring for Linux, or Keychain for macOS).

- For machine-to-machine authentication or browser-based authentication in fallback mode, you must manually define the passkey using an environment variable and update it regularly.

If you delete the credentials file, any authentication IDs you set up before will be lost. A new credentials file will be created when you set up your account again.

Read the following topics for more information:

- [Browser-Based Authentication Standard Process and Fallback Mode](#)
- [Machine-to-Machine Authentication \(CI/CD\)](#)
- [Using Machine-to-Machine Authentication for CI/CD Environments](#)
- [Execution Context for Secure Credentials Storage](#)
- [Setting Up Environment Variables for SuiteCloud SDK Tools](#)
- [Troubleshooting Guide for SuiteCloud SDK](#)

Watch the following help video for information about Secure Credentials Storage for SuiteCloud SDK:

 [Secure Credentials Storage for SuiteCloud SDK](#)

Browser-Based Authentication Standard Process and Fallback Mode

When you use browser-based authentication, SuiteCloud SDK creates the credentials file (`credentials_browser_based.p12`) and encrypts it with an auto-generated passkey. The passkey is stored in your machine's secure storage and is updated regularly when the token expires.

When you do operations that require NetSuite authentication, SuiteCloud SDK retrieves the passkey from secure storage and uses it to decrypt the credentials file. For this reason, ensure that secure storage is accessible and properly configured on your machine if you want to use browser-based authentication.

If secure storage isn't available, you can define a fallback passkey by creating the `SUITECLOUD_FALLBACK_PASSKEY` environment variable. This passkey must contain 32 to 100 alphanumeric characters. For information about how to set environment variables for different operating systems, see [Setting Up Environment Variables for SuiteCloud SDK Tools](#).



Important: The fallback passkey is a temporary solution for when secure storage isn't available, is locked, or isn't properly configured. If you can't use secure storage, you can keep using the fallback passkey for browser-based authentication.

In this case, changing the passkey is your responsibility, and it should be done regularly for optimal security. Note that when you change the passkey, the current credentials file will no longer work. After changing the passkey, you must delete the old credentials file and set up your account again.

When secure storage is available in your machine, ensure to remove the fallback passkey environment variable. Otherwise, you'll get an error.

Machine-to-Machine Authentication (CI/CD)

Machine-to-machine authentication uses two environment variables:

- **SUITECLOUD_CI:** Set this to **1** (preferred) or anything other than zero to run operations in a CI/CD environment using machine-to-machine authentication.
- **SUITECLOUD_CI_PASSKEY:** Set the passkey you'll use to create the credentials file. The passkey needs 32 to 100 alphanumeric characters and should be updated regularly.



Important: When you change the passkey, your current credentials file won't work anymore. After changing it, you'll need to delete the old credentials file and set up your account again.

To set up and manage authentication IDs for machine-to-machine authentication, you only need to set the SUITECLOUD_CI_PASSKEY environment variable. To use the authentication IDs to run operations that need authentication, you need to set both the SUITECLOUD_CI_PASSKEY and SUITECLOUD_CI environment variables.

When you use machine-to-machine authentication, SuiteCloud SDK creates the credentials file (credentials_ci.p12) and encrypts it with the passkey you set in the SUITECLOUD_CI_PASSKEY environment variable. When you do operations that require NetSuite authentication, SuiteCloud SDK retrieves the passkey from the environment variable and uses it to decrypt the credentials file.

To use your authentication credentials in your CI/CD environment, here's what you need to do:

- Upload the credentials_ci.p12 file to your CI/CD environment securely.
- Set the SUITECLOUD_CI_PASSKEY environment variable in your CI/CD environment and use the same passkey you used to create the credentials file.
- Set the SUITECLOUD_CI environment variable to **1** in your CI/CD environment to enable machine-to-machine authentication.

With the credentials and environment variables set, SuiteCloud SDK tools can do operations that need NetSuite authentication in the CI/CD pipeline.

For more information, see [Using Machine-to-Machine Authentication for CI/CD Environments](#).

For information about setting environment variables for different operating systems, see [Setting Up Environment Variables for SuiteCloud SDK Tools](#).

Using Machine-to-Machine Authentication for CI/CD Environments

Machine-to-machine authentication is a robust and efficient way to automate your CI/CD processes securely. To set it up, you need to configure environment variables and credentials. SuiteCloud SDK tools then handle secure authentication in your CI/CD pipeline, ensuring automated and protected processes.



Note: You need two environment variables for machine-to-machine authentication:

SUITECLOUD_CI: This variable enables the machine-to-machine authentication mechanism. Set it to 1 or any non-zero value to activate it.

SUITECLOUD_CI_PASSKEY: This variable stores the passkey for your authentication credentials. Be sure to use a strong, alphanumeric string with 32-100 characters, and update it regularly for better security.

For more information, see [Setting Up Environment Variables for SuiteCloud SDK Tools](#).

To be authenticated using machine-to-machine authentication:

1. Generate a secure passkey that meets the required character requirements.
2. Set the SUITECLOUD_CI_PASSKEY environment variable to the passkey you generated.
3. Run `account:setup:ci` locally to authenticate the accounts you need for deployment and validation in the CI/CD environment.

Note: Be sure to save the auth ID—you'll need it later.

SuiteCloud SDK creates the `credentials_ci.p12` file in the user's local home directory at: `~/suitecloud-sdk/credentials_ci.p12`. This file stores the auth ID (that is, your credentials) and is encrypted with the passkey from Step 1.

Note: Do the next steps in your pipeline.

4. Transfer the credentials securely. Upload the `credentials_ci.p12` file to your CI/CD environment and place it in the CI/CD user's home directory: `~/suitecloud-sdk/credentials_ci.p12`. You might need to create the `.suitecloud-sdk` folder if it doesn't exist.
5. Set the SUITECLOUD_CI environment variable to 1 in your CI/CD environment to enable the machine-to-machine authentication.
6. Ensure the SUITECLOUD_CI_PASSKEY environment variable in your CI/CD environment matches the passkey you used to encrypt the credentials.

To deploy or validate your project, ensure the auth ID you use matches the one in your credentials file from Step 3.

When you use SuiteCloud CLI for Node.js, ensure the project in your CI/CD pipeline uses the auth ID you created during the authentication setup (Step 3). The default auth ID for deployment or validation can't be changed through SuiteCloud CLI for Node.js in a pipeline. You may need to manually update the `DefaultAuthID` parameter in your `project.json` file. For example:

```
1 | project.json
2 | {
3 |   "defaultAuthId": "locallyCreatedAuthID"
4 | }
```

Note: Update the passkey regularly. Changing the passkey will invalidate your current credentials file, so you'll need to regenerate the credentials with the new passkey.

Execution Context for Secure Credentials Storage

The execution context in SuiteCloud SDK controls how authentication and credential management work. It is used by the SuiteCloud SDK tools to determine the following:

- Which authentication option you're allowed to use—either browser-based or machine-to-machine authentication.
- The credentials file used to store authentication details and do operations that require authentication
- The passkey used to encrypt and decrypt the credentials file

Note: Auth IDs created in the browser-based context aren't accessible in the machine-to-machine context, and inversely. So, an auth ID made in one context can't be used in the other.

The environment variables on your machine and whether secure storage is available determine the execution context. You can set up one or more of these environment variables—or none, if you want to use the default context.

- **SUITECLOUD_FALLBACK_PASSKEY:** Stores the passkey to encrypt and decrypt the `credentials_browser_based.p12` file when you're using browser-based authentication in fallback mode.
- **SUITECLOUD_CI:** Used to indicate that you want to run operations in a CI/CD environment using machine-to-machine authentication.
- **SUITECLOUD_CI_PASSKEY:** Stores the passkey to encrypt and decrypt the `credentials_ci.p12` file when you're using machine-to-machine authentication.

The table below shows the conditions for each execution context and gives more details. If your setup doesn't match any of these, it's considered an invalid execution context.

Execution Context	Environment variable configuration and secure storage status	Description
Browser-based (default)	Secure storage is available. Environment variables configuration: ■ <code>SUITECLOUD_FALLBACK_PASSKEY</code> is not set ■ <code>SUITECLOUD_CI_PASSKEY</code> is not set ■ <code>SUITECLOUD_CI</code> is not set	■ It allows browser-based authentication. ■ This is the default execution context, meant for use on a user's machine. ■ Uses the <code>credentials_browser_based.p12</code> credentials file. ■ It uses the auto-generated passkey stored in your machine's secure storage to unlock the credentials file. This passkey is automatically refreshed when the token changes. ■ It allows all other operations, except for machine-to-machine authentication.
Browser-based fallback	Secure storage is unavailable. Environment variables configuration: ■ <code>SUITECLOUD_FALLBACK_PASSKEY</code> is set to a value between 32 and 100 alphanumeric characters. ■ <code>SUITECLOUD_CI_PASSKEY</code> is not set ■ <code>SUITECLOUD_CI</code> is not set	■ It allows browser-based authentication. ■ This execution context is for when secure storage isn't available on a user's machine. ■ Uses the <code>credentials_browser_based.p12</code> credentials file. ■ It uses the passkey defined in the <code>SUITECLOUD_FALLBACK_PASSKEY</code> environment variable to unlock the credentials file. This passkey isn't automatically refreshed and must be manually updated. ■ It allows all other operations, except for machine-to-machine authentication.
Machine-to-machine authentication setup	Secure storage status is not relevant in this case. Environment variables configuration: ■ <code>SUITECLOUD_FALLBACK_PASSKEY</code> is not set ■ <code>SUITECLOUD_CI_PASSKEY</code> is set to a value between 32 and 100 alphanumeric characters. ■ <code>SUITECLOUD_CI</code> is not set	■ It allows machine-to-machine authentication. ■ This execution context can be used to set up and manage authentication IDs for a CI/CD environment. ■ It uses the <code>credentials_ci.p12</code> credentials file. ■ It uses the passkey defined in the <code>SUITECLOUD_CI_PASSKEY</code> environment variable to unlock the credentials file. This passkey isn't refreshed automatically and has to be updated manually. ■ It only allows you to set up a new auth ID or select an existing one. Other commands (such as <code>deploy</code>, <code>validate</code>, etc.) will not work.

Execution Context	Environment variable configuration and secure storage status	Description
		Browser-based authentication is not allowed in this setup.
Machine-to-machine	Secure storage status is not relevant in this case. Environment variables configuration: - SUITECLOUD_FALLBACK_PASSKEY is not set - SUITECLOUD_CI_PASSKEY is set to a value between 32 and 100 alphanumeric characters. - SUITECLOUD_CI is set to 1	- It allows machine-to-machine authentication. - This execution context is meant for use in a CI/CD environment. - It uses the credentials_ci.p12 credentials file. - It uses the passkey defined in the SUITECLOUD_CI_PASSKEY environment variable to unlock the credentials file. This passkey is not automatically refreshed and must be manually updated. - It allows all other operations (such as deploy, validate, etc.), except for browser-based authentication.

For more information, see [Setting Up Environment Variables for SuiteCloud SDK Tools](#).

Setting Up Environment Variables for SuiteCloud SDK Tools

You can set up the following environment variables for SuiteCloud SDK:

- SUITECLOUD_FALLBACK_PASSKEY
- SUITECLOUD_CI
- SUITECLOUD_CI_PASSKEY
- SUITECLOUD_PROXY (use this only if you're working behind a VPN connection and you can't access netsuite.com)

Below are some ways you can set environment variables on Windows, macOS, and Linux.

To set the environment variables on Windows:

- Click the Windows icon and type **environment variables**.
- In the results list, click **Edit the system environment variables**.
- In the System Properties dialog, click **Environment Variables**.
- Under System variables, click **New**.
- In the New System Variable dialog, enter the variable name and value, then click **OK**.
- Restart your terminal or application.

To set the environment variables on macOS:

- Open the terminal.
- Mac uses different shell profiles based on shell type. If you already know which shell you're using, skip to Step 3.

To find out what shell you are using, run the following command:

```
1 | echo $SHELL
```

- If the output contains `/bin/zsh`, you're using **zsh**.
- If the output contains `/bin/bash`, you're using **bash**.

3. Run the appropriate command based on the shell profile you are using:

- For zsh:

```
1 | nano ~/.zprofile
```

- For bash:

```
1 | nano ~/.bash_profile
```

4. Add the environment variable to the file in this format:

```
1 | export VARIABLE_NAME=value
```

For example:

```
1 | export SUITECLOUD_CI_PASSKEY=0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

5. Save your changes and close the file. In nano:

1. Press Ctrl + X to exit.
2. Press Y to confirm changes.
3. Press Enter to save.

6. To apply the changes immediately, run the following command:

- For zsh:

```
1 | source ~/.zprofile
```

- For bash:

```
1 | source ~/.bash_profile
```

7. (Optional) To ensure that the variable is set correctly, check its value with this command:

```
1 | echo $VARIABLE_NAME
```

To set the environment variables on Linux:

1. Open the terminal.
2. Run the following command:

```
1 | gedit ~/.bashrc
```

3. Add the environment variable to the file in this format:

```
1 | export VARIABLE_NAME=value
```

For example:

```
1 | export SUITECLOUD_CI_PASSKEY=0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ
```


4. Save your changes and close the file.
5. To apply the changes immediately, run the following command:

```
1 | source ~/.bashrc
```

6. (Optional) To ensure that the variable is set correctly, check its value with this command:

```
1 | echo $VARIABLE_NAME
```

Troubleshooting Guide for SuiteCloud SDK

This topic provides information about errors related to SuiteCloud SDK and how to fix them.

- [Authentication Errors](#)
- [Credentials and Passkey Errors](#)
- [Execution Context Errors](#)

Authentication Errors

Error Message	Scenario	Solution
Secure storage is inaccessible. Ensure that the secure storage in your system is properly configured and accessible.	The OS secure storage isn't accessible. It might be locked or not set up right.	Check the OS-based secure storage service on your machine. ■ For Windows, ensure that Credentials Manager is working. ■ For MacOS, ensure that Keychain is working. ■ For Linux, make sure the default keyring is available and unlocked. You can use Seahorse or another tool that lets you see the GNOME Keyring v2.22+/Libsecret. As a temporary fix, while the secure storage issue is being resolved, you can use the <code>SUITECLOUD_FALLBACK_PASSKEY</code> environment variable to set the passkey for the credentials file. If the <code>credentials_browser_based.p12</code> file already exists and the passkey from secure storage cannot be retrieved, delete this file and authenticate again to run operations that require authentication. After you re-authenticate, the new credentials file will be saved with the new passkey you set in <code>SUITECLOUD_FALLBACK_PASSKEY</code>. For more information, see Browser-Based Authentication Standard Process and Fallback Mode and Setting Up Environment Variables for SuiteCloud SDK Tools.
Secure storage is inaccessible. Ensure that the secure storage in your system is	Secure storage isn't accessible in a CI/CD environment. Actions that need credentials (like deploy) fail with this error because	Ensure that the following environment variables are set: ■ <code>SUITECLOUD_CI=1</code>

Error Message	Scenario	Solution
properly configured and accessible.	the right environment variables aren't set up to use the credentials in the credentials_ci.p12 file.	SUITECLOUD_CI_PASSKEY set to an alphanumeric string with 32-100 characters. With these environment variables set, actions that need credentials (like deploy) can access the auth IDs stored in the credentials_ci.p12 file. For more information, see Execution Context for Secure Credentials Storage.
There was an error with the private key used to authenticate. The private key does not match the certificate ID <certificateId>.	When you're using machine-to-machine authentication, the private key and certificate ID you entered don't match.	Ensure that you're using the right private key and certificate ID together.
There was an error with the private key used to authenticate. Verify the contents of the private key.	When you're using machine-to-machine authentication, the private key file you specified is either corrupted or not valid.	Generate a new certificate. For information about how to obtain the certificate ID, see OAuth 2.0 Authentication for SuiteCloud SDK .
There was an error with the certificate ID used to authenticate. Verify that certificate ID <certificateId> is valid and not expired or revoked in account <accountId>.	When you're using machine-to-machine authentication, the certificate ID you specified is not valid.	Ensure that OAuth 2.0 is enabled in your account. For more information, see the help topic Enable the OAuth 2.0 Feature. Verify that the certificate ID is valid. In your NetSuite account, go to Setup > Integration > Manage Authentication > OAuth 2.0 Client Credentials (M2M) Setup.

Credentials and Passkey Errors

Error Message	Scenario	Solution
The current passkey cannot decrypt the credentials file <credentialsFile>. Try deleting the file and setting up your account again.	The current passkey isn't the same as the one used to create the existing credentials file.	If the passkey is set in the environment variable SUITECLOUD_FALLBACK_PASSKEY or SUITECLOUD_CI_PASSKEY, ensure that you've got the right passkey. If you don't have the passkey for the existing credentials file, delete the credentials file from C:\Users\<username>\.suitecloud-sdk, then set up your account again with any SuiteCloud SDK tool.  Note: When you delete the credentials file, any authentication IDs you set up before will be lost. When you re-authenticate, a new credentials file will be generated with the correct passkey. If the passkey is set in the SUITECLOUD_FALLBACK_PASSKEY or SUITECLOUD_CI_PASSKEY environment variable, that passkey will be used to encrypt the new credentials file. Otherwise, if you use browser-based authentication, a

Error Message	Scenario	Solution
		new passkey will be generated and stored in the secure storage.
The passkey required to decrypt your credentials is missing. Try deleting the credentials file <credentialsFile> and setting up your account again.	The passkey is missing or there is a problem with the Oracle NetSuite: Account Credentials entry in your OS secure storage.	Delete the credentials file from the C:\Users\<username>\.suitecloud-sdk folder, then set up your account again using any of the SuiteCloud SDK tools.  Note: When you delete the credentials file, any authentication IDs you set up before will be lost. When you re-authenticate, a new credentials file and passkey will be generated, and the passkey will be stored in the secure storage.
Credentials passkey could not be retrieved from the secure storage. Try deleting the current credentials file <credentialsFile> and setting up your account again.	There was a problem getting the passkey from secure storage.	Delete the credentials file from the C:\Users\<username>\.suitecloud-sdk folder, then set up your account again using any of the SuiteCloud SDK tools.  Note: When you delete the credentials file, any authentication IDs you set up before will be lost.
The credentials passkey set in the environment variable <environmentVariable> must be between 32 and 100 alphanumeric characters.	The passkey you set in the environment variable doesn't meet the requirements.	Use a passkey with 32 to 100 alphanumeric characters.
The credentials passkey set in secure storage must be between 32 and 100 alphanumeric characters.	The passkey stored in secure storage doesn't meet the requirements.	Delete the credentials file from the C:\Users\<username>\.suitecloud-sdk folder, then set up your account again using any of the SuiteCloud SDK tools.  Note: When you delete the credentials file, any authentication IDs you set up before will be lost.

Execution Context Errors

Error Message	Scenario	Solution
The following environment variables are currently set: <environmentVariables>. The operation you are trying to perform is not allowed with the current environment variable configuration.	Your current setup doesn't match any valid execution context. For example, both SUITECLOUD_CI_PASSKEY and SUITECLOUD_FALLBACK_PASSKEY environment variables are set.	Check your environment variable setup for the execution context you want to use, and make sure it's set up right. For more information, see the Execution Context for Secure Credentials Storage table.
In the current execution context, you can perform only machine-to-machine authentication or operations that do not require authentication.	Only the SUITECLOUD_CI_PASSKEY environment variable is set, and you're trying to do operations that require NetSuite authentication.	If you want to use machine-to-machine authentication (CI/CD), set the SUITECLOUD_CI environment variable to 1. If you want to use browser-based authentication, remove the SUITECLOUD_CI_PASSKEY environment variable.

Error Message	Scenario	Solution
Machine-to-machine authentication is not allowed for the current execution context.	You're trying to run machine-to-machine authentication, but the required environment variables aren't set up right.	Ensure that the you've set these environment variables: ■ SUITECLOUD_CI=1 ■ SUITECLOUD_CI_PASSKEY (must contain 32 to 100 alphanumeric characters)
Browser-based authentication is not allowed for the current execution context.	The SUITECLOUD_CI_PASSKEY environment variable is set, and you're trying to run browser-based authentication.	Remove the SUITECLOUD_CI_PASSKEY environment variable.
Secure storage is available, but the SUITECLOUD_FALLBACK_PASSKEY is set. Remove the fallback passkey to use secure storage.	The SUITECLOUD_FALLBACK_PASSKEY environment was set when secure storage wasn't available, but it wasn't removed after you fixed secure storage.	Remove the SUITECLOUD_FALLBACK_PASSKEY environment variable.